Supporting Information

New routes for spermine biosynthesis

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Table S1. Proteins analyzed in this study

Species (Phylum)	**Protein** [GenBank acc. no.]	Size (a.a.)
***Paracoccus denitrificans* (α-Proteobacteria)**	**CASDH** [WP_011749229]	407
***Paracoccus denitrificans* (α-Proteobacteria)**	**CASDC** [WP_011749230]	368
***Agrobacterium tumefaciens* (α-Proteobacteria)**	**CASDH** [WP_010973612]	412
***Agrobacterium tumefaciens* (α-Proteobacteria)**	**CASDC** [WP_010973611]	365
***Brucella abortus* (α-Proteobacteria)**	**CASDH** [WP_002963500]	413
***Brucella abortus* (α-Proteobacteria)**	**CASDC** [WP_002963499]	365
***Deferribacter desulfuricans* (Deferribacterota)**	**CAPADH** [WP_013008001]	406
***Deferribacter desulfuricans* (Deferribacterota)**	**CAPADC** [WP_013008000]	394
***Deferribacter desulfuricans* (Deferribacterota)**	**APT** [WP_013008000]	277
***Clostridium leptum* (Bacillota)**	**CAPADH** [EDO61991]	399
***Clostridium leptum* (Bacillota)**	**CAPADC** [EDO61992]	376
***Clostridium leptum* (Bacillota)**	**APT** [EDO61989]	284
***Hydrogenimonas thermophila* (ε-Proteobacteria)**	**CAPADH** [WP_317066177]	399
***Hydrogenimonas thermophila* (ε-Proteobacteria)**	**CAPADC** [WP_317066176]	378
***Psychromonas marina* (γ-Proteobacteria)**	**CANSDH** [WP_284203352]	405
***Psychromonas marina* (γ-Proteobacteria)**	**CANSDC** [WP_284203351]	373
***Sporomusa ovata* (Bacillota)**	**APT** [WP_021166903]	278
***Leptotrichia buccalis* (Fusobacteriota)**	**APT** [WP_015770029]	280
***Haliangium ochraceum* (Myxococcota)**	**APT** [WP_012829793]	283

CASDH, carboxyspermidine dehydrogenase; CASDC, carboxyspermidine decarboxylase; CAPADH, carboxyaminopropylagmatine dehydrogenase; CAPADC, carboxyaminopropylagmatine decarboxylase; CANSDH, carboxynorspermidine dehydrogenase; CANSDC, carboxynorspermidine decarboxylase: APT, aminopropyltransferase.

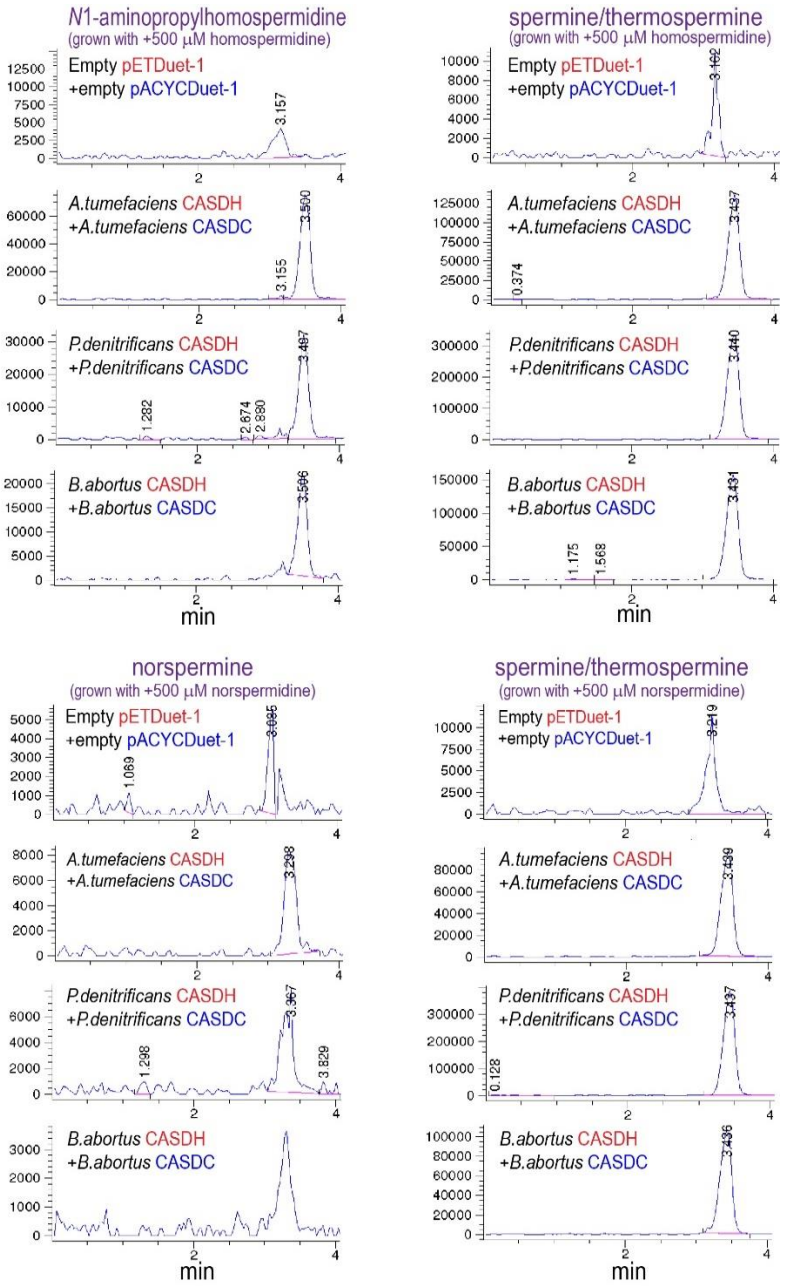

Figure S1. N^1-aminopropylhomospermidine and norspermine production by CASDH/CASDC in *E. coli* BL21*speD*. Spermidine-void *E.coli* BL21*speD* strains coexpressing CASDH/CASDC were grown with either 500 μM homospermidine or 500 μM norspermidine. Polyamines from cell extracts were benzoylated and analyzed by LC-MS. Shown are the Extracted Ion Chromatograms (EICs) for tetrabenzoylated N^1-aminopropylhomospermidine (mass tolerance window 633.04:634.04), tetrabenzoylated norspermine (605:606) and tetrabenzoylated spermine/thermospermine (619.02:620.02). CASDH and CASDC from the indicated species were coexpressed from pETDuet-1 and pACYCDuet-1, respectively, in spermidine-void *E.coli* BL21*speD*. The y-axis represents arbitrary units of ion intensity, and all samples were grown, extracted and analyzed together.

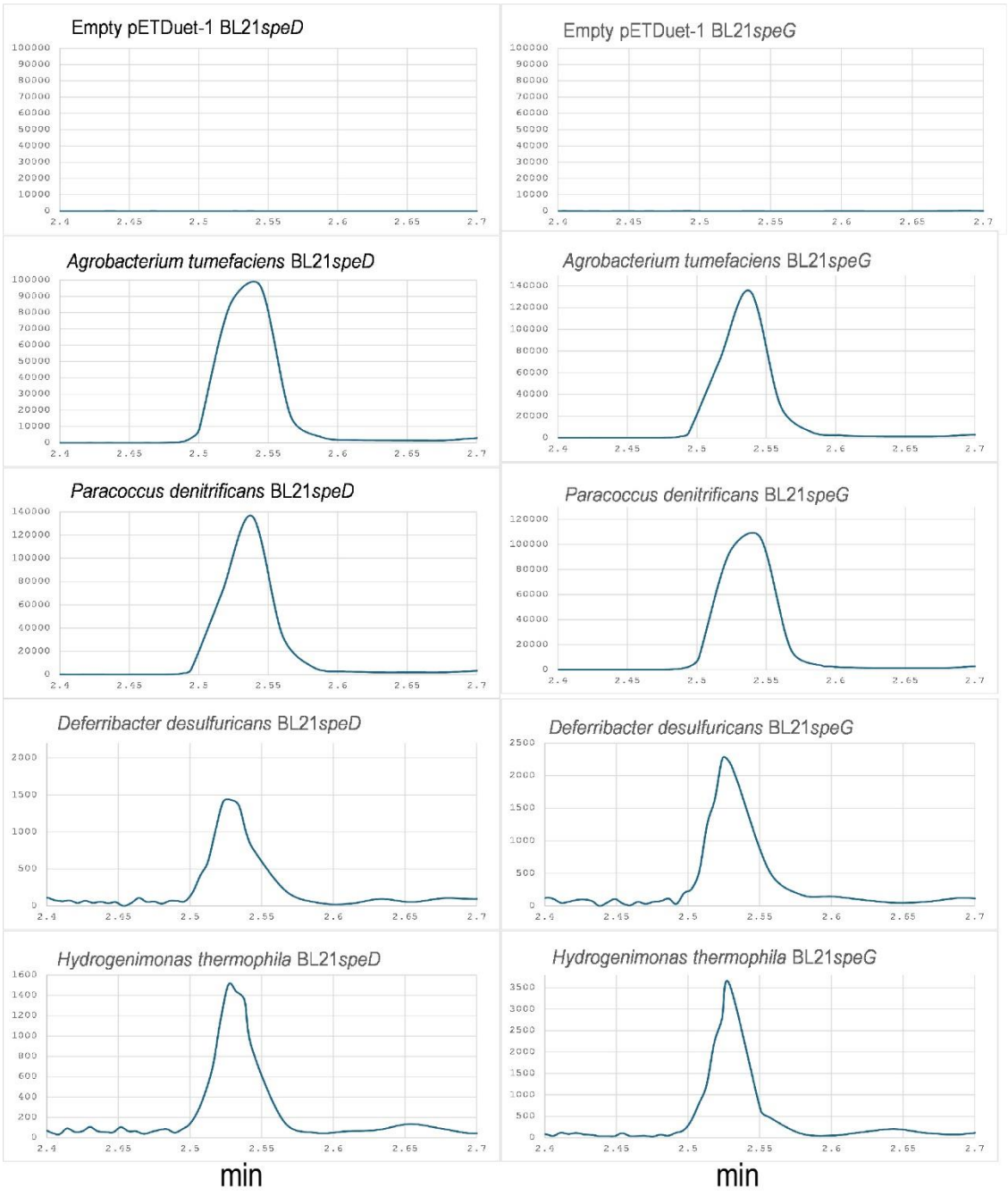

Figure S2. High Resolution-LCMS detection of underivatized carboxyspermidine after expression of CASDH homologs in *E. coli* BL21*speD* and BL21*speG*. Shown are the HR-LCMS chromatograms relating to Table 2 in the main text.

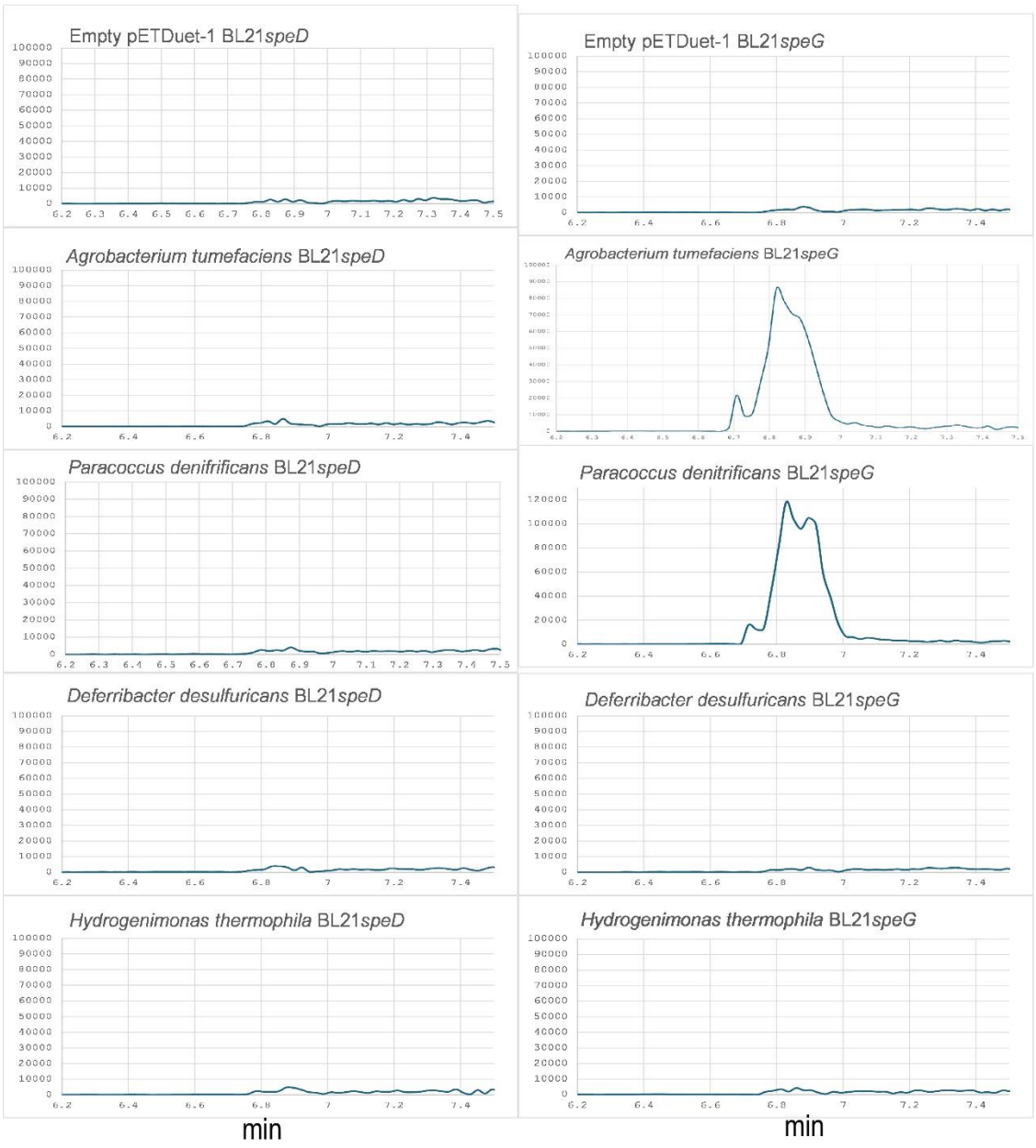

Figure S3. High Resolution-LCMS detection of underivatized carboxyspermine/ carboxythermospermine after expression of CASDH homologs in *E. coli* BL21*speD* and BL21*speG*. Shown are the HR-LCMS chromatograms relating to Table 3 in the main text. Note that carboxyspermine and carboxythermospermine are not clearly resolved.